

Introduction

To Cloud Fundamentals

- Early Predictions on Cloud Computing
- What is Cloud Computing?
 - The Definition(s)
- Business Drivers
 - Why Cloud Computing?
- Basic Concepts and Terminology
- Cloud Characteristics
- Cloud Delivery Models
- Cloud Deployment Models

Early Predictions on Cloud Computing

John McCarthy

Computer Scientist, publicly proposed in 1961

“If computers of the kind I have advocated become the computers of the future, then computing may someday be organized as a public utility just as the telephone system is a public utility”

Leonard Kleinrock

Chief Scientist of ARPANET project, 1969 (ARPANET - Advanced Research Project Agency Network)

“As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, we will probably see the spread of computer utilities”

- **Gartner** (in one of the reports on Strategic Technology & Industry Trend, 2008)

“A style of computing in which scalable and elastic IT-enabled capabilities are delivered as a service to external customers using internet technologies.”

- **Forrester Research** (@2009)

“A standardized IT capability (services, software, or infrastructure) delivered via internet technologies in a pay-per-use, self-service way.”

- **NIST** (National Institute of Standards and Technology, USA, 2011)

“Cloud computing is a model for enabling ubiquitous, convenient, on-demand access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

Business Drivers (why use Cloud Computing?)

- Capacity Planning
 - Determine and fulfil future demands of an organization's IT resources, products and services
 - Over-provisioning – costly and resource drain
 - Under-provisioning – inefficient and frustrating end user experience
- Cost Reduction
 - Data centre maintenance
 - Utility bills
 - Capital expenses for power and cooling
 - Technical and Operational staff
 - Routine activities
 - upgrades and patches
 - testing and deployment cycles
 - Security and Access control measures to protect resources
 - Track S/W license renewals, support arrangements (AMC), etc.
- BCP/DR
 - RTO – Recovery Time Objective
 - RPO – Recovery Point Objective
- Organization Agility
 - Responsiveness to change
 - Address Competitive pressure
 - Strategic business goal

- Cloud
 - Distinct Distributed IT environment that is designed for the purpose of remotely provisioning scalable and measured IT resources
 - Privately owned and offers access (via internet) to IT resources that is metered
- IT Resource
 - Physical
 - Physical Server
 - Network device
 - Virtual IT-related artefact
 - Virtual Server
 - Custom software program
- On-Premise
 - IT-resources hosted in a conventional IT data centre of an organizational boundary
 - Managed and secured by organization
 - May not be accessible to staff or public outside organization without adequate authorization

- Cloud Provider
 - Party that provides cloud-based IT resources
- Cloud Consumer
 - Party that uses cloud-based IT resources
 - Organizations generally have cloud provisioning contract with Cloud Providers
- Scaling
 - IT-resource scaling has two types, to handle increase or decrease in demand (usage fluctuations)
 - Horizontal Scaling – allocating or releasing same type of IT-resources
 - Scaling out & Scaling in
 - Vertical Scaling – replacing by higher or lower capacity IT-resources. **Requires down time.**
 - Scaling up and Scaling down

- IT environment requires a specific set of characteristics to enable the remote provisioning of scalable and measured IT resources in an effective manner
 - On-demand usage
 - Cloud consumer to self-provision required cloud based IT resources as and when needed
 - Cloud provider to measure the usage and charge accordingly
 - Ubiquitous access
 - Ability of a cloud based service or IT resource to be widely accessible with support from range of devices, transport protocol, interfaces and security technologies
 - Web Interface (Desktop, mobile, TV, etc.), API, CLI.
 - Multitenancy
 - Cloud provider pools its IT resources to serve multiple cloud service consumers
 - Dynamically assigned and reassigned according to cloud service consumer demands
 - Pools – Physical servers, virtual servers, storage, network, CPU, memory, etc.

- Elasticity
 - Core justification for cloud computing adoption
 - Ability to transparently scale IT resources as required in response to runtime conditions pre-defined by cloud consumer
 - Cost savings realized
- Measured usage
 - Ability of a cloud platform to keep track of the usage of its IT resources by cloud consumer and cloud provider both
 - Usage reporting also helps cloud consumer to fine tune/re-configure IT resources to meet business demands
- Resiliency
 - Addresses reliability and availability of organization applications via
 - Fault tolerance or
 - Failover technique readily available
 - BCP/DR inbuilt in the cloud platform

- **Infrastructure as a Service (IaaS)**

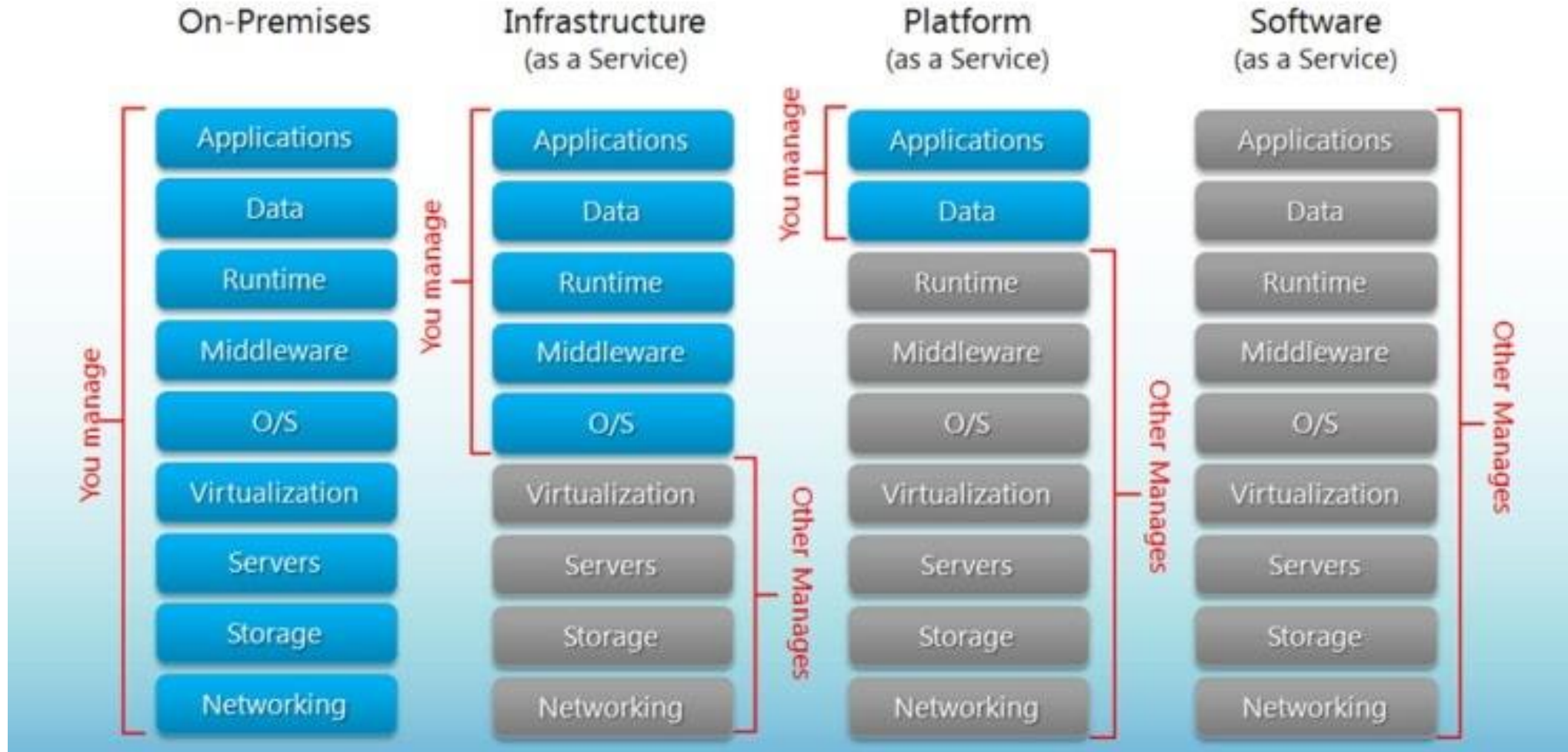
- Self contained IT environment comprised of infrastructure-centric IT resources managed via cloud service-based interfaces and tools
- Consumer has
 - Full administrative control
 - Full access to virtualized infrastructure-related IT resources &
 - At times full access to underlying physical IT resources

- **Platform as a Service (PaaS)**

- Pre-packaged ready-to-use products/service
- Consumer would invest in PaaS environments
 - Extend on-premise environment into the cloud for scalability and economic purpose
 - Use ready-made environment to substitute an on-premise environment
 - Wants to become a cloud provider to offer cloud based services to other external cloud consumers

- **Software as a Service (SaaS)**

- Reusable cloud service widely available to a range of cloud consumers
- Products can be leased or used for different purposes via subscription or contractual terms



<https://www.bigcommerce.com/blog/saas-vs-paas-vs-iaas/#the-three-types-of-cloud-computing-service-models-explained>

Other Cloud Delivery Models (specialized variations in use today)

- Storage as a Service
- Database as a Service
- Security as a Service
- Communication as a Service
- Integration as a Service
- Testing as a Service
- Process as a Service

- **Public Cloud**

- Publicly accessible cloud environment owned by a third-party cloud provider

- **Community Cloud**

- Multi-tenant platform which allows several companies work on the same platform, given that they have similar needs and concerns
 - Security model testing
 - Public cloud environment simulation and testing specific security or regulatory concerns

- **Private** requiring access to this environment assume cloud consumer role

- **Hybrid CloudCloud**

- Owned by a single organization
 - Specific department plays cloud provider role
 - Other department
- A cloud environment having combination of two or more cloud deployment models

